Individual Assessment Coversheet

To be attached to the front of the assessment.

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Group: Group 2

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Indicate	Yes	No
Plagiarism report attached		X

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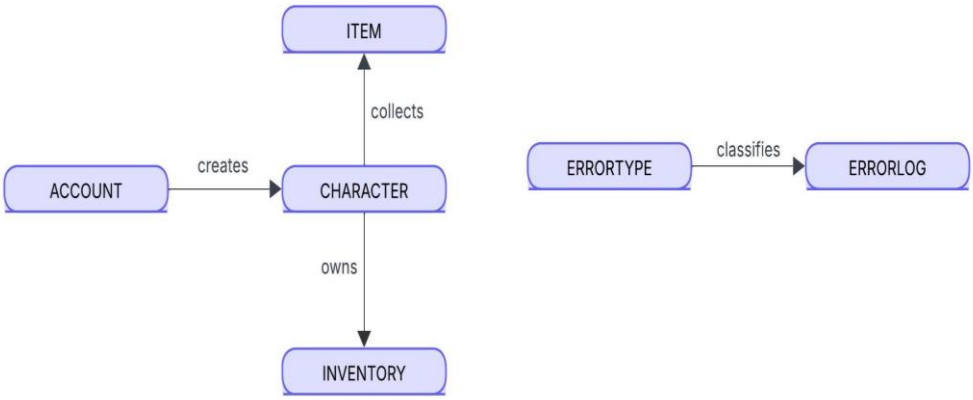
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Question 1

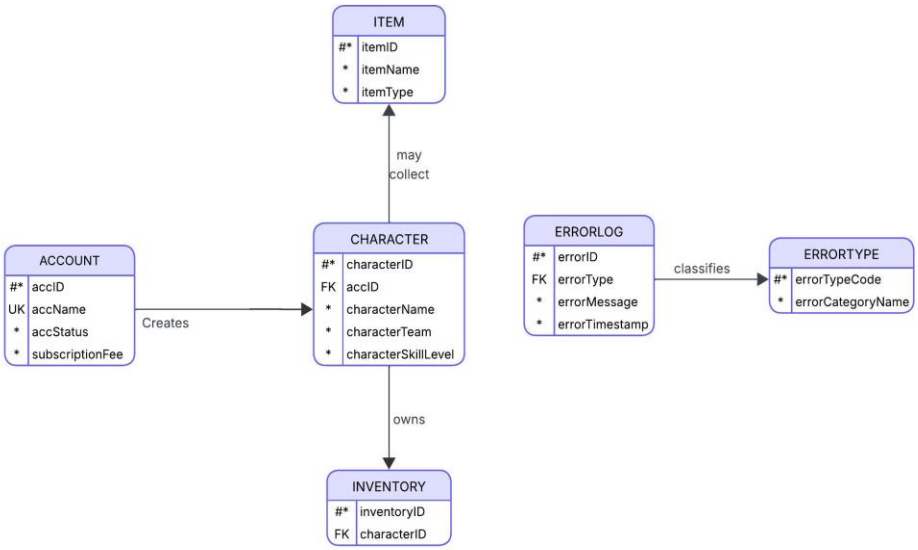
1.



Notes:

- An ER diagram is a type of flow chart that show how entities relate to each other in a system
- An entity is an object or concept that represents important data. They are typically nouns
- I am not including Players and an entity because all the data or attributes of players can be entered into Account
- The entities are:
  ACCOUNT (stores user details and payments), CHARACTER (stores the characters that are created by the players), ITEM (are collected by characters), INVENTORY (stores the items collected), ERRORLOG (can be logged by anyone or anything and the scenario does not say so they are not connected to character or account) and ERRORTYPE (they are classified by ERRORLOG)

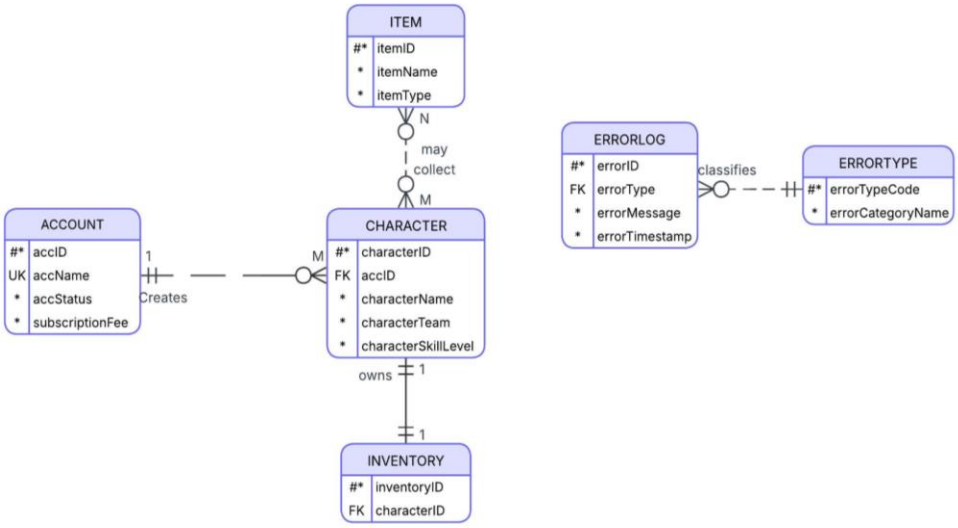
1.2



Notes:

- An attribute is a characteristic of an entity
- An UID (#) is an attribute or a set of attributes that uniquely identifies the instance of the entity. It is usually named ID attribute in the entity.
- FK (Foreign Key) is an UID that comes from another entity and identifies the relationship between the two

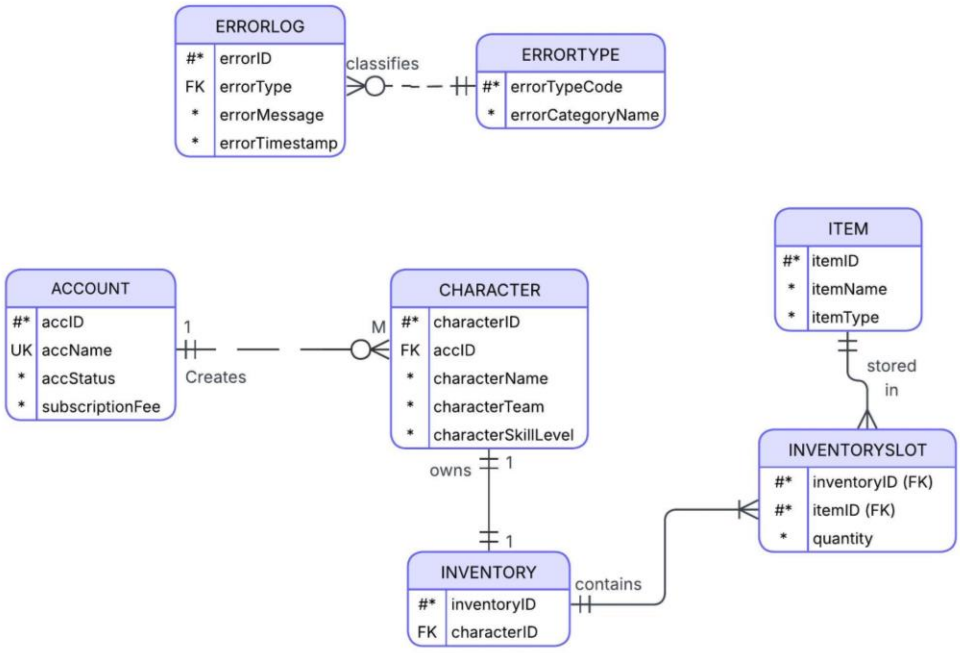
1.3



Notes:

- Cardinality defines the numerical attributes for relationships between entities (one-to-one; many-to-one; or many-to-many)
- Optionality checks if something is mandatory or optional
- On my diagram:
    - ACCOUNT may create one or more CHARACTER. CHARACTER must belong to one ACCOUNT
    - CHARACTER may collect zero or more ITEM. ITEM may be collected by zero or more CHARACTER.
    - ERRORTYPE may classify zero or more ERRORLOG. ERRORLOG must be categorized under only one ERRORTYPE
    - CHARACTER must own only one INVENTORY. INVENTORY must be owned by only one CHARACTER
- || indicates only one; * indicates mandatory; --- indicates optionality; E indicates many; 0 represents optionality

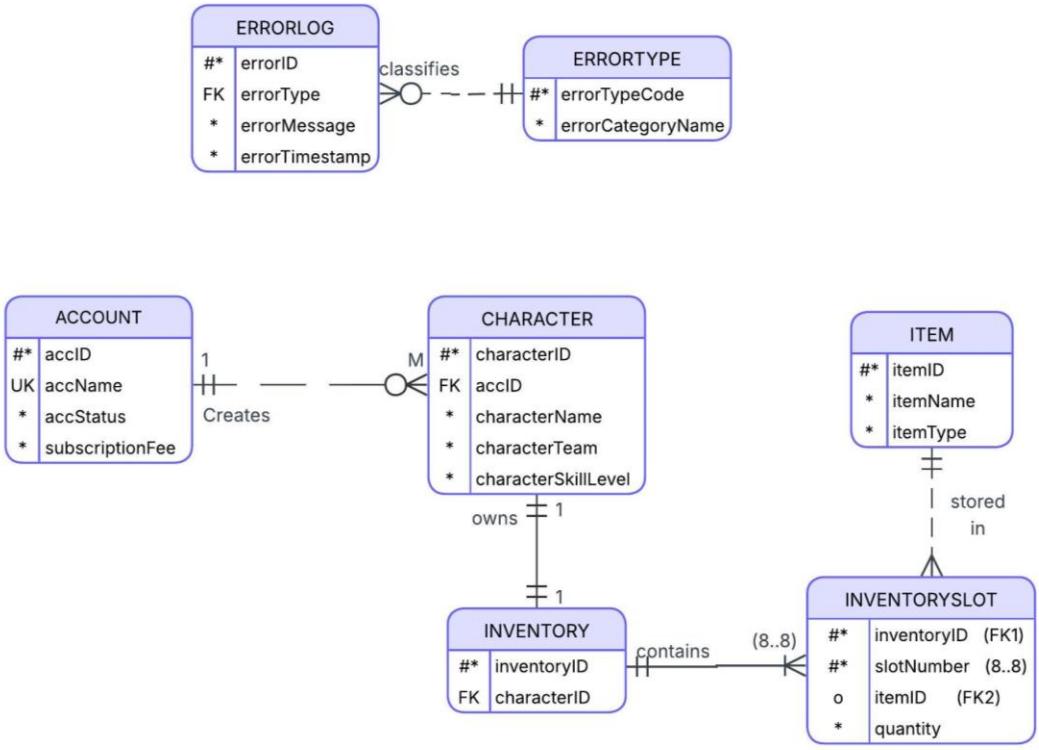
1.4



Notes:

- Many CHARACTER can collect many ITEM this will cause duplication and make querying complex and knowing which item is owned by which character.
- A many-to-many is solved by having an intersection entity that connects the two entities. In this case, the intersection entity is INVENTORYSLOT because each slot is uniquely identified by a composite UID inherited from both parent IDs (CHARACTER and ITEM).
- The composite UID for the intersection entity is inventoryID (from INVENTORY) and itemID (from ITEM).
- Even though the many-to-many relationship exists between CHARACTER and ITEM, a character does not hold the items directly; they hold them in an inventory in a specific slot. That is why INVENTORYSLOT connects ITEM and INVENTORY nor CHARACTER and ITEM directly.

1.5

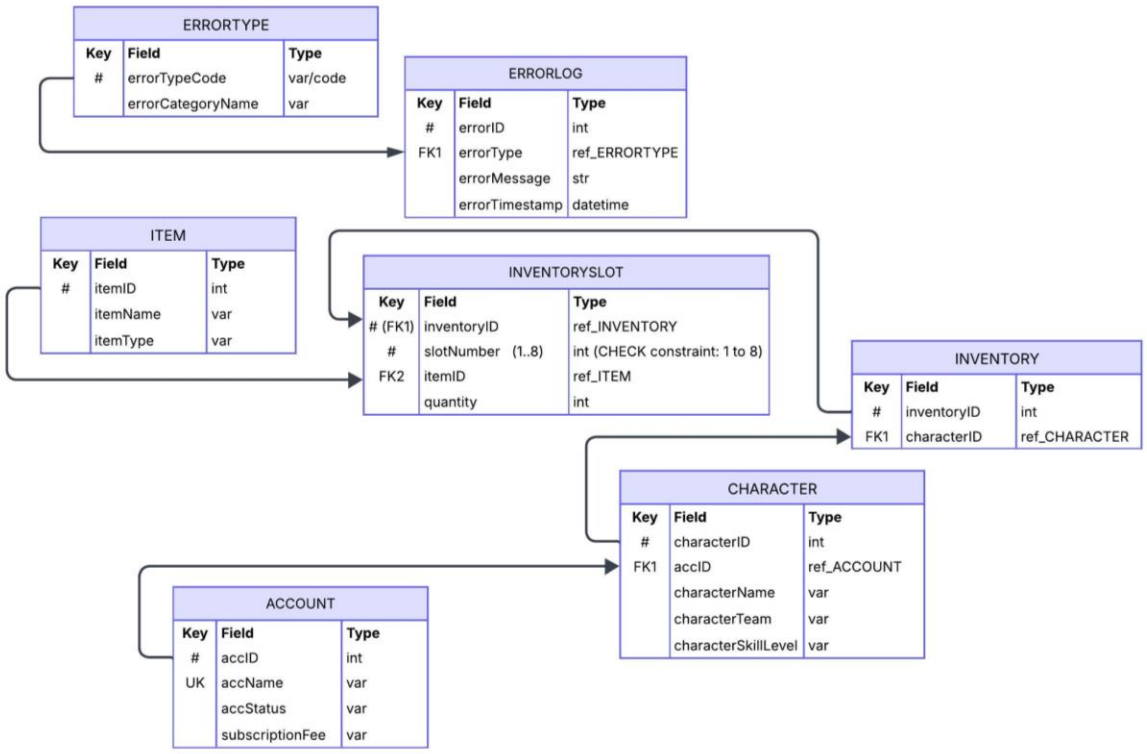


Notes:

- The 8..8 notation enforces that every inventory must have exactly 8 slots. On the slotNumber attribute, it enforces that slot numbers are restricted to values 1 to 8, which ensures that each slot is uniquely numbered. Together they make sure that the inventory has exactly 8 slots.
- ItemID is optional because the scenario does not say that all slots must be occupied at all times, so optional itemID allows empty slots.
- SlotNumber is an attribute because it represents the physical position of a slot inside an inventory. It shows which item occupies which position in the inventory.
- slotNumber and inventoryID are both primary keys (composite PK) because a slot identity is defined by which inventory it belongs to and which position it occupies. One without the other is not unique because a single inventoryID can appear in multiple slots, and a single slotNumber can appear in every inventory.

Question 2

2.1



Notes

- A schema diagram is a visual that shows how entities, attributes, and relationships in a database are connected.
- My schema diagram shows key, field (attribute) and type.
- The types are integer (int) and varchar (var).
- ref_X (x representing anything) means it references the type from the Entity that key comes from; it will have the same data type as the one in the entity comes from.
- According to relational database theory, every attribute must belong to a specific domain (set of valid values). A data type like int is too broad, so the CHECK constraint is restricting that domain. So, by writing the CHECK constraint 1 to 8, i am matching the int domain to restrict up to 8 slots.

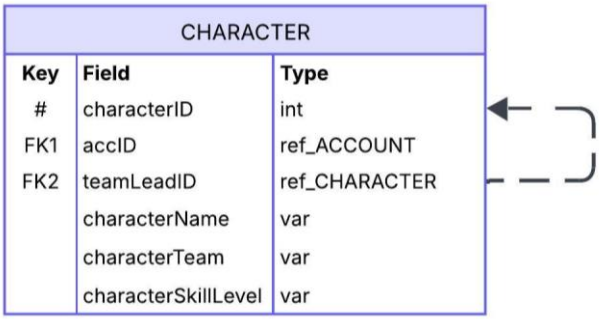
2.2

INVENTORYSLOT		
Key	**Field**	**Type**
# (FK1)	inventoryID	ref_INVENTORY
#	slotNumber (1..8)	int (CHECK constraint: 1 to 8)
FK2	itemID	ref_ITEM
	quantity	int

Notes:

- INVENTORYSLOT is the intersection table. Its composite Primary Key is inventoryID and slotNumber, they cannot be on their own because neither is unique, but when they are together, the form a PK. This is created to solve the many-to-may relationship between CHARACTER and ITEM.
- InventoryID (FK1) from INVENTORY, acts as both a part of the PK and FK1 because as FK1 it links back to INVENTORY table to verify which slot it belongs to.
- itemID (FK2) from ITEM, act as a normal FK that links back to ITWM to specify what item is inside a slot. It is optional because it can be NULL.
- The attribute quantity is used to track how many items are combined inside a slot.
- A composite Primary Key is a type of key that uses a relationship with another entity that can be used to form part of its UID.

2.3



Notes:

- A recursive relationship is one between one instance of an entity and another instance of the dame entity. In this case, characters form a team, and those teams have a leader or leaders who are also a character in the database.
- A relationship that is recursive will most likely be optional on both ends.

2.4

ACCOUNT		
Key	**Field**	**Type**
#	accID	int
UK	accName	var
	accStatus	var
	subscriptionFee	var

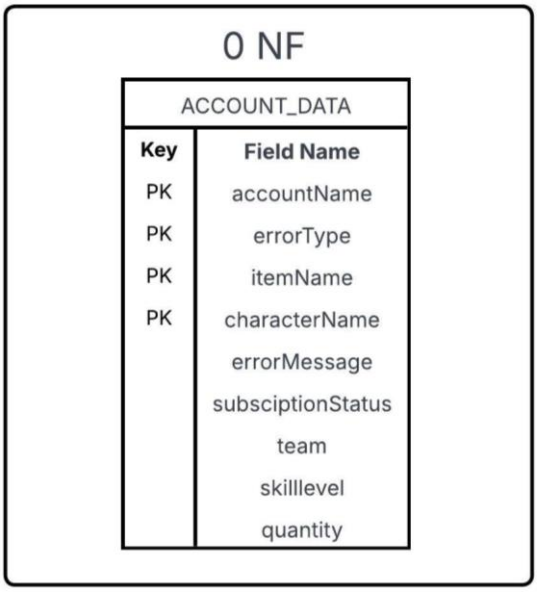
A Unique Key exists on the ACCOUNT able as accName

Notes:

- A UK are columns that uniquely identify items in the erd, they are optional so they can be a NULL value.
- A PK is any combination of attributes and/or relationships that uniquely identifies an instance of an entity. They are mandatory, so they can never be NULL.
- accName is a unique key because it identifies a user. It is unique to prevent two players from having the same account name.
- accName is not the primary key because primary keys usually do not change and a player can change their account name.

Question 3

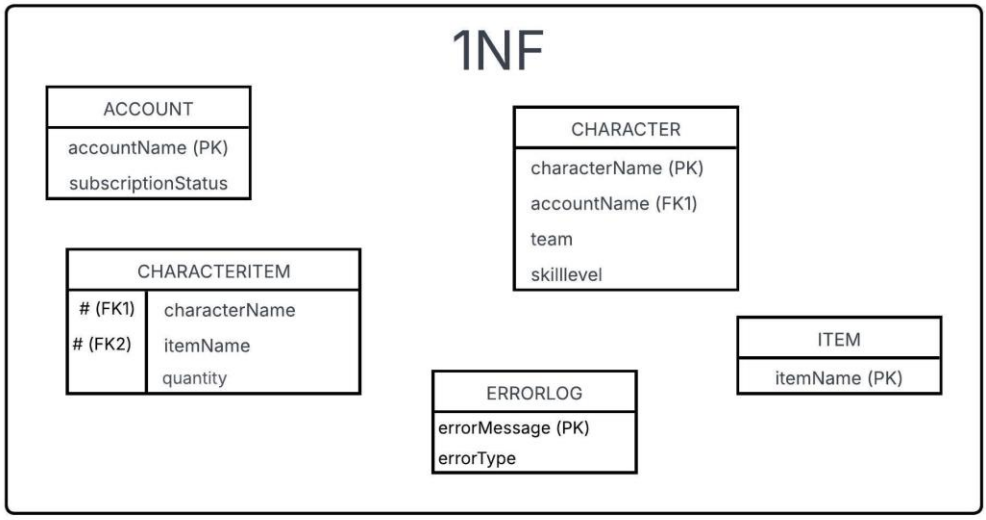
3.1



Notes:

- A normal form is a step or stage during the normalisation process. Each normal form has its own rules, which is why each form results in the data being separated into more logically related data groups.
- Normalisation is important when designing a large database since following the set of steps is required, and it is less common for human error to happen.
- 0 NF is when all the data is in one list, and a PK is found for the whole list.
- My diagram has composite primary keys because the data contains multiple nested repeating groups because account details, character statistics, inventory slots, and error logs are stored in one file. A single attribute would violate entity integrity so a Composite Primary Key (accountName, characterName, itemName and errorType) will guarantee unique identification

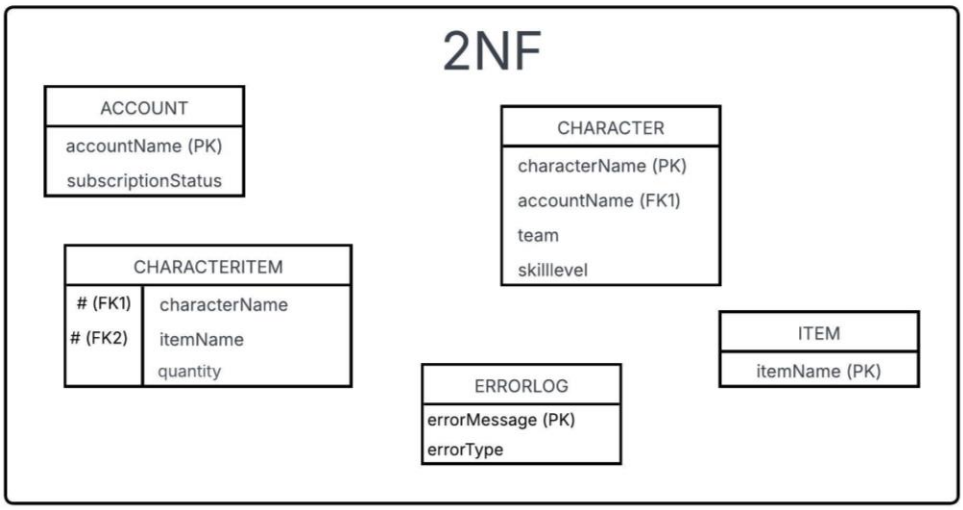
3.2



Notes:

- The point of 1NF is to remove alll groups that are repeating from 0NF.
- A repeating group is a group of attributes that could repeat more than once for the selected PK.
- The data is separated into individual entities and attributes. This makes sure there are no repeating values.
- The repeating groups were removed by creating separate entities (ACCOUNT, CHARACTER, ITEM and ERRORLOG).
- Foreign Keys are introdused to link related entities
- This improves how the data is organized

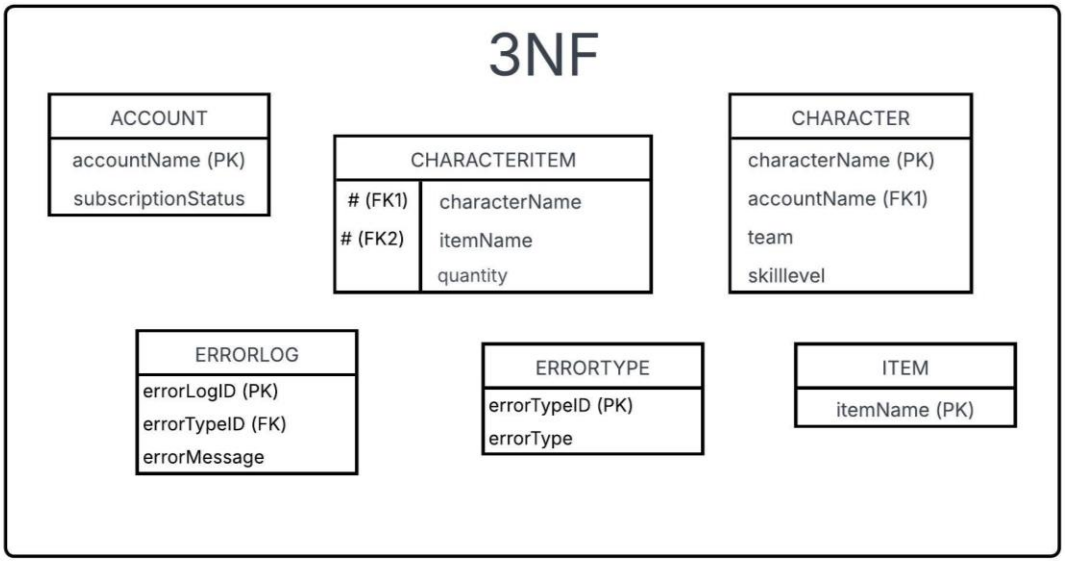
3.3



Notes:

- The goal in 2NF is to remove groups that are not dependent on the whole composite Primary Key, meaning they can exist without PK
- Partial dependencies are removed
- Attributes that depend on only a part of the composite key are moved as their own entities. Character details are placed into CHARACTER entity and item details in ITEM entity
- This ensures that every non-key attribute depends on the whole primary key.
- In this scenario there is not much difference between 1NF and 2NF

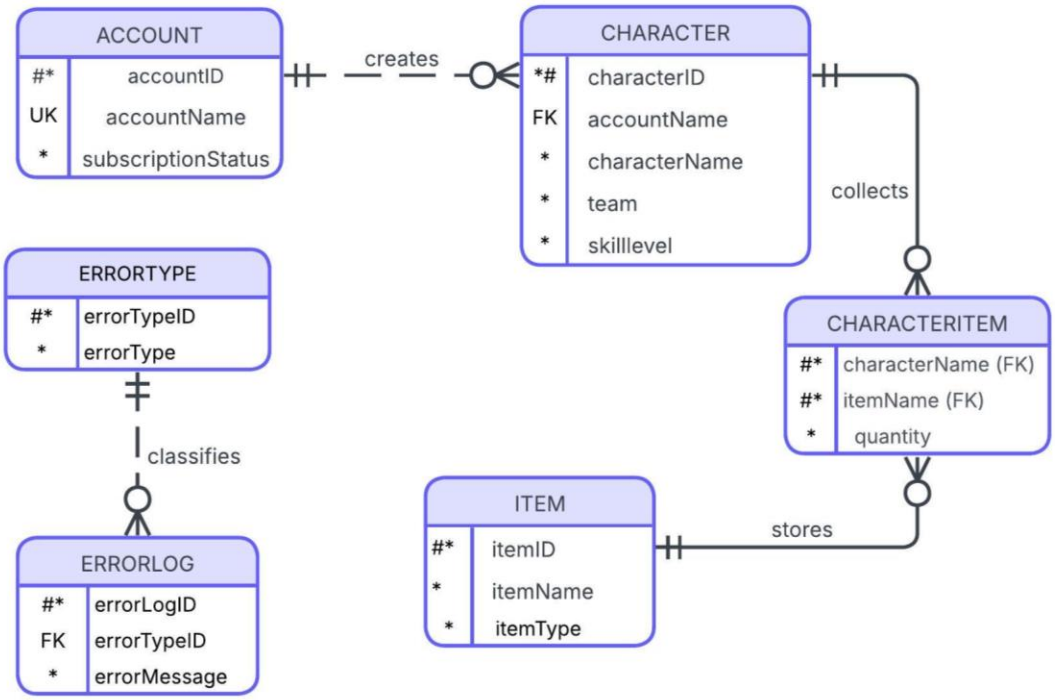
3.4



Notes:

- In 3NF no attribute in a group should depend on another attribute that is not the PK.
- It identifies and removes all non-key attribues that are mutually dependent.
- It creates one-to-many relationships.
- Error type information was separated into its own ERRORTYPE entity, and ERRORLOG references it through a relationship.

3.5

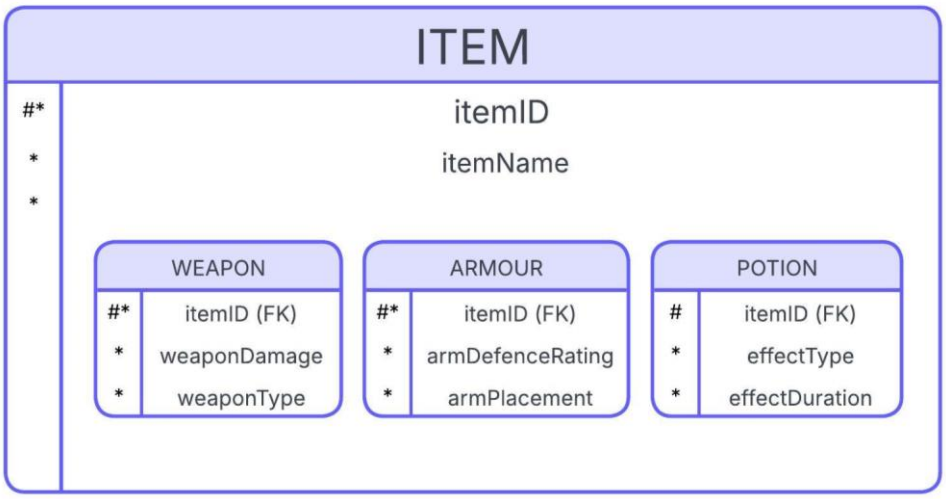


Notes:

- ACCOUNT is mandatory (||) and CHARACTER needs to belong to exactly one account, but ACCOUNT can create (----) 0 or more CHARACTER (0<).
- CHARACTER is mandatory (||) and CHARACTERITEM needs to belong to a specific character, but CHARACTER can hold 0 or more CHARACTER_ITEM. A solid line because the identifying key is inherited as the PK
- ITEM is mandatory (||) and CHARACTIREITEM must reference a valid item, but ITEM can be in 0 or more CHARACTER_ITEM (0<)
- ERRORTYPE is mandatory (||) and ERROTLOG must be classifies under a valid system error code, but ERRORTYPE can categorise 0 or more EROORLOG (0<). The --- line because it is a non-identifying relationship

Question 4

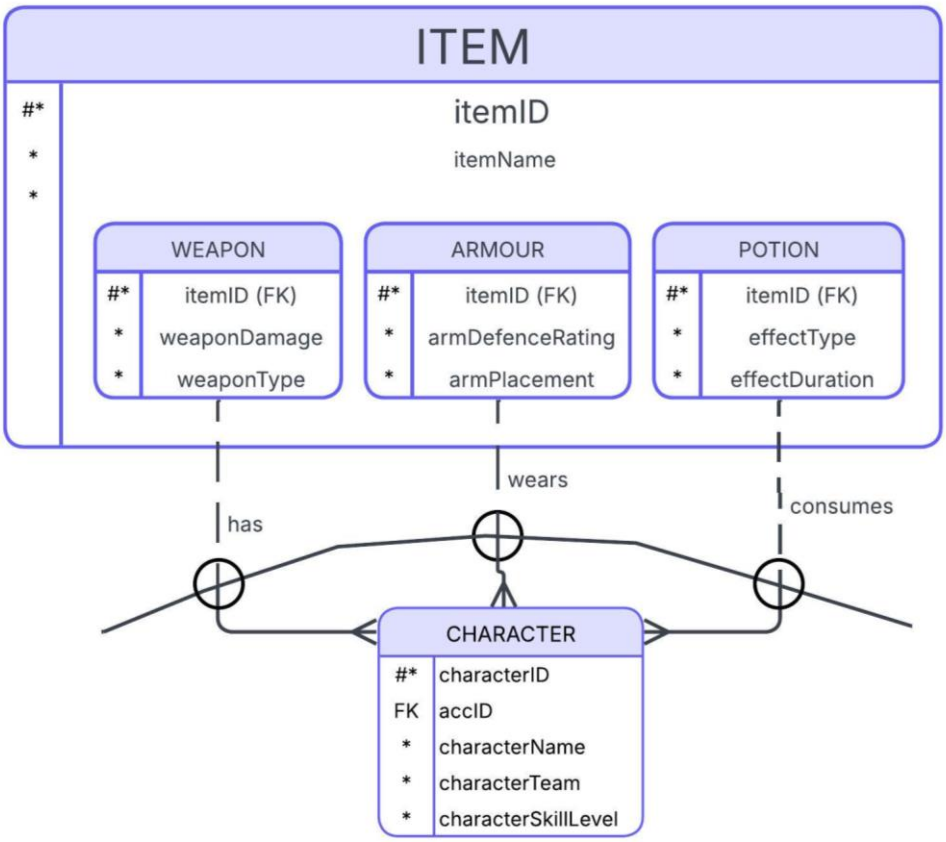
4.1



Notes:

- A supertype can be seen as a container entity, which contains smaller groups that are mutually exclusive and have their own attributes.
- A subtype is a smaller group inside a supertype which inherits all the attributes of the supertype. They are mutually exclusive, meaning that an instance of one subtype cannot be an instance of another type.
- When i googled how to show inheritance in subtypes and super types, the examples used a small triangle written is a, that connects all entities using lines, but i decided to use the example on myLMS which looks like my diagram.
- Each supertype can have its own attributes and relationships that are unique to it, and the UID is placed in the supertype.
- ITEM is the supertype, and WEAPON, ARMOUR, and POTION are the subtypes. They all inherit the PK from the supertype, and I included it in the subtypes as FK.

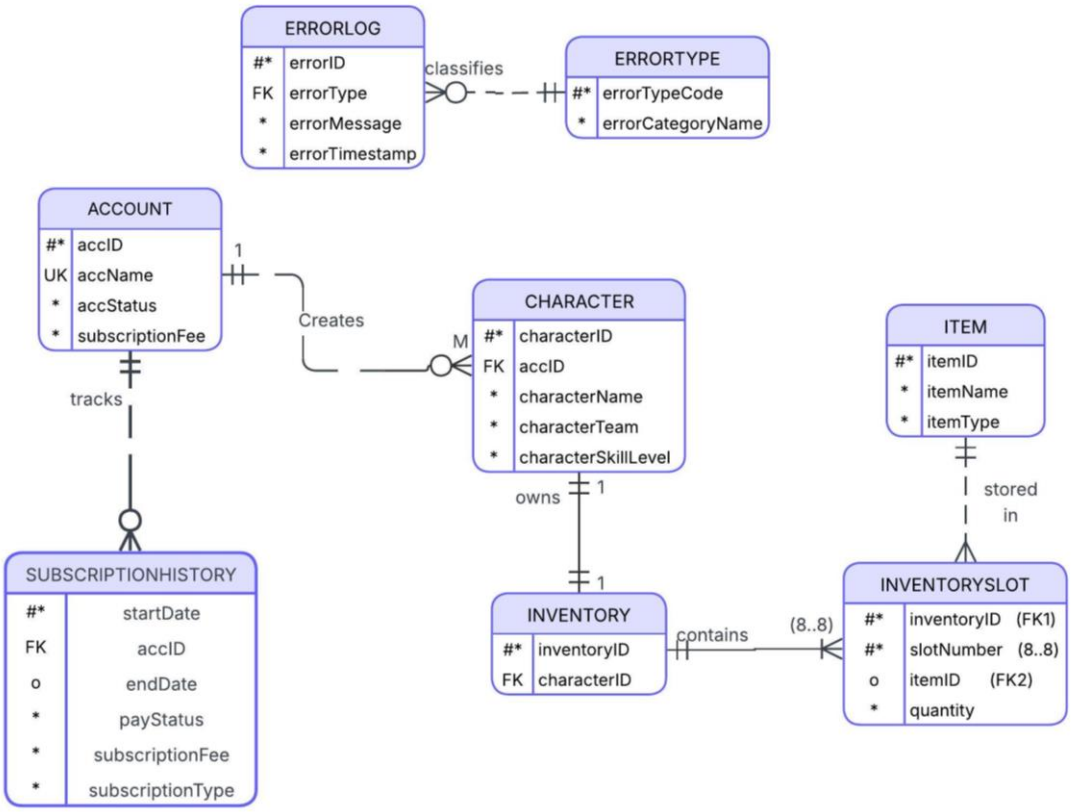
4.2



Notes:

- When an entity can have a relationship with only one relationship at a time with a subtype, the exclusive relationship is modelled using an arc.
- An arc needs to have the same optionality on both ends.
- The small circle at the spot where the arc and relationship meet shows that the arch includes the relationship.
- In this case, a character can only activate one item at a time.

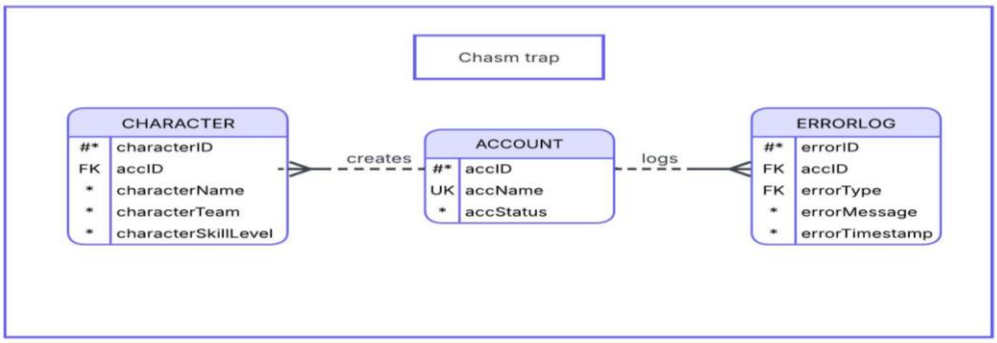
4.3

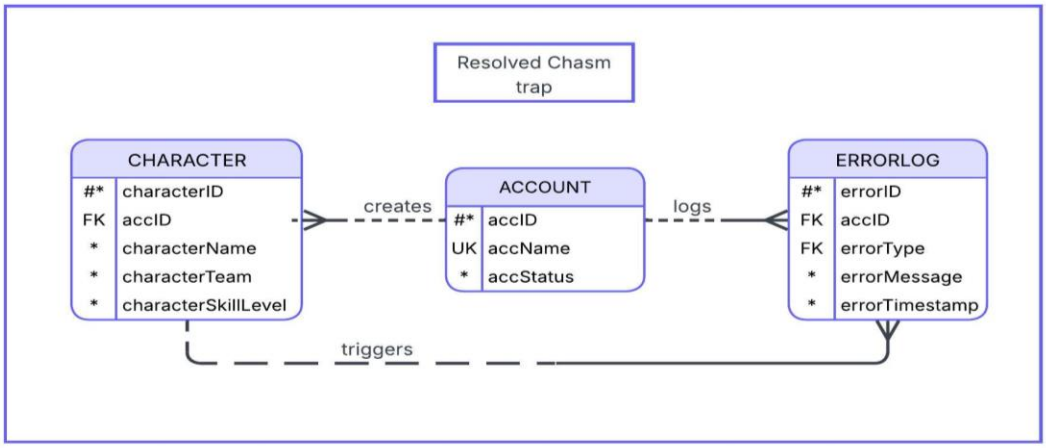


Notes:

- Certain entities have historical data that needs to be stored. When that happens, extra entities and relationships need to be added to the diagram because the data for the entity that is involved will almost certainly repeat.
- This requires extra storage, so it is costly.
- So, in this case, if a user pays in August and cancels in September, and decides to join again in December, that data might be lost. So, to track this change, a historical entity is created.
- To be able to track the change, the primary key needs to cobine the account's identity with a time component. In my case, accID links the entity to ACCOUNT entity, the startDate is the PK, the endDate is optional because a user's subscription can be active, paystatus tracks if the payment was successful or not, the subscriptionFee tracks price change over time, and then subscriptionType checks whether it is a monthly, yearly or quarterly subscription.

4.4





Notes:

- A connection trap is when a certain combination of entities can cause the sata the is retrieved from the ER diagram or database system to be misinterpreted.
- A fan trap is when there is a many-to-many relationship between 3 or more entities. A chasm trap is when there are missing relationships in the ER diagram, thus means that information can be lost when an instance that links to another instance is lost.
- In this case, we have a chasm trap because if a player decides to delete one of their old game characters, or a critical system crash happens on the log in screen before the player selects a character, the link is lost. The database will have no way to attach ERRORLOG to the parent ACCOUNT, causing data to be lost.
- To fix this, ERRORLOG needs to have a relationship with both CHARACTER and ACCOUNT at the same time. This makes sure that even if character records change, or are not there, the root account occurrence is linked to the data.

Conclusion

The process of creating an ER diagram has taught the importance of correctly identifying entities, relationships, and cardinality before adding detail. Key decisions included leaving ERRORLOG standalone since the scenario provided no information about what generates errors and linking the CHARACTER to ITEM relationship through INVENTORY and INVENTORYSLOT rather than directly, because items are held in specific slots with specific quantities, not by characters directly. This taught me to think outside the box and to think logically without losing the core of the business problem.

Normalisation showed how a single flat table containing repeated data can be systematically broken down into clean, linked tables by removing repeating groups, partial dependencies, and transitive dependencies, and this reduces redundancy and making the database easier to maintain. My lecturer taught us the last term first, so i got to reverse engineer how we got there. Now i can take any messy data and turn it into ERD.

Advanced modelling concepts like supertype/subtype structures, arcs, historical entities, and relationship traps demonstrated that real database design goes beyond basic ERDs. They showed that there is more to ERDs than diagrams, they solve problems. Identifying and correcting fan traps and chasm traps ensures that data remains accurate, and queries return complete results.

Overall, this project showed that good database design is not just technical; every decision must be justified by the business requirements of the system being modeled. I feel more confident in my abilities after this project. The module did not have activities, so we did not do many hands on, but this project solved that, it wrapped everything we covered in this course

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